Date
30/11/24

Metals and Alloys

* **Metals**:- The elements which are hard, dense, ductile, malleable, good conductors of heat and electricity, electropositive in nature and posses shining lustre and high melting and boiling points are called metals.
For eg- Copper, iron, gold etc.

* **Non-Metals**:- The elements which are soft, brittle, less dense, dull in appearance, poor conductors of heat and electricity, electronegative in nature and posses low melting and boiling points are called non-metals.
For eg- phosphorus, Sulphur, Oxygen etc.

* **Metalloids**:- The elements that show the properties of both metals and non-metals are known as metalloids.
For eg- Antimony, Arsenic, Bismuth etc.

* **Occurrence of metals**:-
Metals occur in nature in two forms which are given below:
(i) In Free Or Native state
(ii) In Combined state
(a) As Minerals (b) As Ores

1. **Native (or free state)**:- The metals which are found in nature in the free or elementary state are called native metals.

For eg- noble metals with low reactivity like silver, gold, platinum etc, occur in native or free state in nature. Lumps of pure metal, found in nature are known as nuggets.

2. <u>Combined state</u>:- When a metal is ~~to~~ found in combination with other elements (mostly non-metals) in nature, it is said to occur in combined state. Most of the metals occur in the combined state in the form of their compounds ~~are~~ as:

(a) <u>Minerals</u>:- Minerals are naturally occuring materials found in the earth's crust. A minerals has a definite chemical composition and structure. It is ~~found~~ formed in nature by inorganic processes. Minerals are generally associated with some unwanted earthy impurities.

(b) <u>Ores</u>:- The minerals from which the metals can be extracted economically and conveniently are called ores.

* <u>Metallurgy</u>:- The branch of science dealing with the methods of extraction of metals from their impurities. Ores and refining them to the purity required for commercial purposes is called metallurgy.

Or,

The extraction of metals in pure state from their ores is called metallurgy.

Or,

The process of extraction of metals from the rich Ores of the metals and their refining is called metallurgy.

* ***Gangue or matrix :-*** The Undesired earthy impurities like sand, rocks, limestone etc. associated with the ~~ore~~ Ore are known as gangue or matrix.

Or,

The worthless earthy impurities like sand, rocks, limestone, clay etc. associated with the Ore are known as gangue or matrix.

These impurities can be acidic (eg- SiO_2, P_2O_5 etc.) or basic (eg- CaO, FeO, $MgCO_3$ etc.)

* ***Flux :-*** The acidic substances like SiO_2, P_2O_5 etc. or basic substances like CaO, MgO etc. which are mixed with the ore to remove acidic or basic impurities are called flux.

Or,

A chemical substance added to convert gangue into fusible mass is known as flux.

Flux combines with the impurities and form a fussed mass which is separated from the metal from time to ~~tim~~ time.

Fluxes are of two types :

(a) Acidic flux : It is used to remove basic impurities like FeO, CaO, MgO etc.. eg - SiO_2, P_2O_5, TiO_2, V_2O_5 etc.

$$\underset{\text{gangue}}{FeO} + \underset{\text{flux}}{SiO_2} \longrightarrow \underset{\text{ferrous silicate slag}}{FeSiO_3}$$

(b) Basic Flux : It is used to remove acidic impurities like SiO_2, P_2O_5 etc. eg - CaO, MgO etc.

$$\underset{\text{gangue}}{SiO_2} + \underset{\text{flux}}{MgO} \longrightarrow \underset{\text{slag}}{MgSiO_3}$$

* Slag :- The fused mass which is separated from the process formed by the combination of flux and impurities is called slag.

Or,

The fusible mass formed by the combination of flux and impurities (matrix) is called slag.

Flux + Matrix ⟶ Slag

Eg - 1. $\underset{\substack{\text{flux} \\ \text{(acidic)}}}{SiO_2} + \underset{\substack{\text{matrix} \\ \text{(Basic)}}}{CaO} \longrightarrow \underset{\text{slag}}{CaSiO_3}$

2. $\underset{\substack{\text{flux} \\ \text{(Basic)}}}{CaO} + \underset{\substack{\text{matrix} \\ \text{(Acidic)}}}{P_2O_5} \longrightarrow \underset{\text{slag}}{Ca(PO_3)_2}$

~~(acidic)~~ ~~(Basic)~~

* Concentration :-

The process of removal of matrix from the crushed ~~ore~~ Ore is Called Concentration.

Or,

The process of increasing the percentage of the material in the ore by removing the gangue is Called Concentration of the Ore.

Method's of Concentration of Ore are as follows :

- Gravity Separation
- Froth Floatation Process
- Electro magnetic Separation
- Liquation.

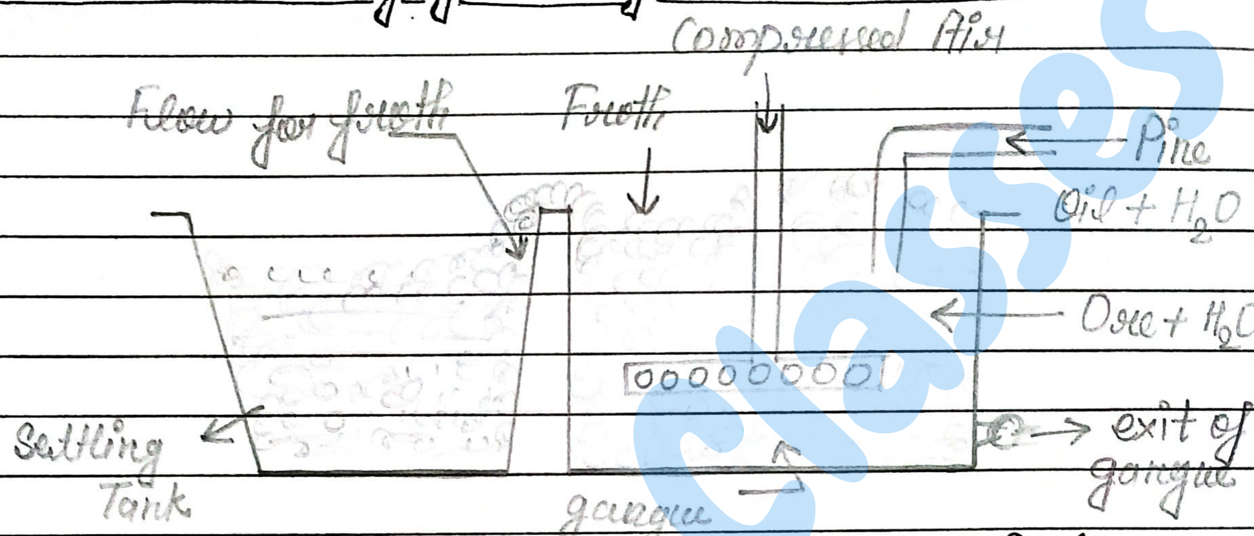
*** Froth Floatation Process :-** This process is used for the Concentration Sulphide Ores. It is used on the fact that Sulphide Ore particles are preferential watted by Oil while the gangue and oxide particles are watted by water Only.

The powdered ore is mixed with little pine Oil and put in a big tank of water. The whole mass is then agitated by passing Compressed air through it. The Ore particles which get preferentially watted by Oil move to the Surface of the form of froth. Whereas, the gangue particles which are watted by water become heavier, slowly sattle down the bottom and are removed from time to time. The froth at the top Carrying Ore particles overflows into the Sattling tank, Where the ore Sattles down after Some time.

* General Metallurgical Operations :-

This ore is worked up for extracting the metal after drying.

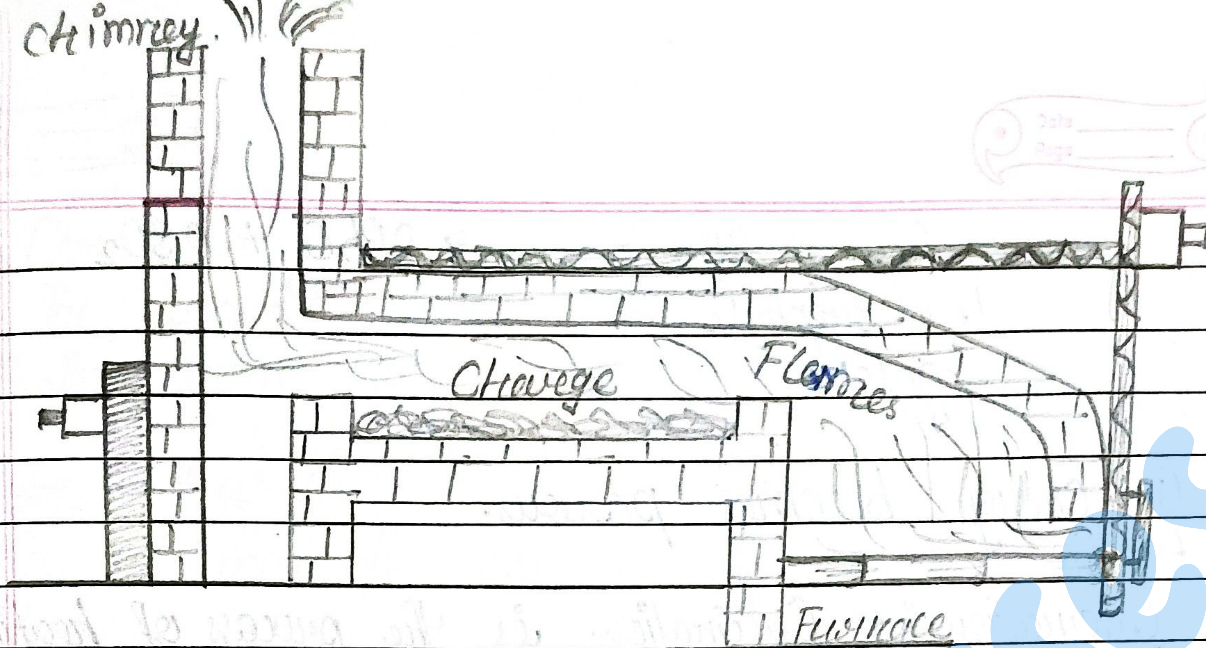
Concentration By froth floatation Process :-



* Conversion of the Concentrated Ore to its Oxide form :-
This Concentrated ore can be Converted into its Oxide from by the following two methods :
(a) Roasting
(b) Calcination.

- Roasting : The process of heating the Ore in excess of air below the fusion point with the ~~process~~ purpose of changing ore into oxide is known as roasting.

Roasting in Reverberatory furnace :-



Roasting is the process of heating the Ore strongly in the presence of excess of ~~ore~~ ____ to a temperature insufficient to melt it. Roasting is done in the hearth of a special type of reverbaratory furnace having a no. of doors which are kept ____ for the free supply of air.

The following changes takes place during roasting:

(i) Moisture is Driven Out.

(ii) Removal of Volatile impurities like Sulphur, arsenic, antimony and phosphorus in the form of their Oxides.

$$S + O_2 \longrightarrow SO_2 \uparrow$$

$$P_4 + 5O_2 \longrightarrow 2P_2O_5 \uparrow$$

$$4As + 3O_2 \longrightarrow 2As_2O_3 \uparrow$$

(iii) The Sulphide Ores undergo Oxidation and changes into their respective Oxides.

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$

Zinc Sulphide
(Zinc blende)

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

Lead sulphide
(galena)

(iv) Charge becomes porous.
~~(v) Ca~~

- Calcination :- Calcination is the process of heating the Concentrated ore ~~strer~~ strongly in the absence or limited supply of air at a temperature below its melting point.

Or,

The process of heating the ore strongly in the absence of air but at a temperature insufficient to melt it is known as Calcination.

Calcination is done in the hearth of a reverberatory furnace when the doors are kept closed. During Calcination the Volatile impurities and moisture are removed and the mass becomes porous.

This process is generally Carried out in the case of Carbonate and hydroxide ______, which looses carbon dioxide and moisture respectively and get Converted into the Oxide of the metal.

The following reactions take place during Calcination.

$$ZnCO_3 \longrightarrow ZnO + CO_2 \uparrow$$
$$FeCO_3 \longrightarrow FeO + CO_2 \uparrow$$
$$Fe_2O_3 \cdot 3H_2O \longrightarrow Fe_2O_3 + 3H_2O \uparrow$$
$$Cu_2CO_3 \cdot Cu(OH)_2 \longrightarrow 2CuO + CO_2 \uparrow + H_2O$$

* Conversion of the Oxide to the metallic form :-
The roasted or the Calcined ore is then Converted to free metal by red-.. which can be done in a no. of ways depending upon the nature of the ______

This may be done by any of the following method.

Smelting (Carbon-reduction method): Smelting is the process of ex-__. a metal by heating the Oxide ore with Carbon.

The roasted ore is mixed with an adequate quantity of Carbon (Coal or --) and heated in a furnace. Carbon reduces the Oxides ore to the free metal.

For eg- $ZnO + C \xrightarrow{\Delta} Zn\uparrow + CO\uparrow$ (vapour)

$SnO_2 + 2C \xrightarrow{\Delta} Sn + 2CO\uparrow$

(molten metal)

$Fe_2O_3 + 3C \xrightarrow{\Delta} 2Fe + 3CO\uparrow$

$Fe_2O_3 + 3CO \xrightarrow{\Delta} 2Fe + 3CO_2\uparrow$

* Properties of Various Steels :-

* Properties of Various Steels :-

1. Low Carbon Steel (Soft Steel) :-

Carbon Content :- 0.05% to 0.20%

Properties :-

- Soft and ductile
- Suitable for welding
- Not Susceptible to heat treatment
- Has low tensile strength

Uses :- Wires, rods, strip plates, boiler tubes etc.

2. Medium Carbon Steel (Mild Steel) :-

Carbon Content :- 0.20% to 0.40%

Properties :-

- Fairly good for welding
- Can be hardened by heat treatment
- Has a good machining property.

Uses :- Railways ~~ax~~ axils, Armature shaft, turbine motors, crank shafts, Fish plates, Cross heads.

3. High Carbon Steel (Hard Steel) :-

Carbon Content :- 0.40% to 0.70%

Properties :-

- Can be welded with a great care.
- Can be imparted a desired hardness by heat treatment.

Uses :- Dies, Sat screws, saws wrenches, wheel for railway Service, Cusion springs, clutch springs etc.

4. Very High Carbon Steel (very hard steel):-
Carbon Content :- 0.70% to 1.50%
Properties :-
- Unweldable
- Can be hardened to a very large extent by heat treatment.

Uses :- Railways rails, plough shovels rock drills, chisels machine tools, knives, springs, ~~hand~~ band saw for cutting steel.

* Give the difference between metals & Non-Metals :-

Metals	Non-Metals
Metals have tendency to loose the electrons.	Non Metals have tendency to accept the electrons.
Ex- Iron, Copper, Chromium, Nickel etc.	Ex- Hydrogen, Carbon, Chlorine, Sulphur etc.
Metals have high melting and boiling point.	Non-Metals have low melting and boiling point.
Metals have Higher density.	Non metals have lower density.
Metals are Solid at Ordinary temperature except mercury which is liquid.	Non metals are Solid, liquid or gas at Ordinary temperature.
Metals are good Conductor of heat and electricity	Non metals are Bad Conductor of heat and electricity.

Metals are malleable and ductile	Non metals are not malleable and ductile.
Metals ~~from~~ form basic Oxides	Non metals form acidic Oxides.
Metals are generally hard.	Non metals are generally soft.

* Differentiate between minerals and ore :-

Minerals	Ore
A naturally Occuring Substances present in earth crust which Contain metal in the free state or in Combined state is Called a mineral	A mineral from which metal can be extracted economically is Called an ore.
Clay and Bauxite both are minerals of Al.	Al can be profitably extracted from bauxite and not from clay. So bauxite is ore of Al. and not clay.
clay is Only mineral.	Bauxite is mineral as well as Ore.

* Distinguish between Calcination and roasting :-

Calcination	Roasting
Ore is heated in the absence of air.	Ore is heated in the presence of air.
This process is used for Carbonates and hydroxide Ores.	This process is used for Sulphide Ores.

• The mass Becomes highly porous.	The mass Becomes less porous.
• It evaporates impurities on heating.	Impurities are Oxidised and then evaporated.
• Decomposition reactions takes place.	Oxidation reactions takes place.

* Mechanical Properties of Metals :-

• Hardness :- Hardness is the ability of the metal to resist wear or abrasion. For eg- Tungsten metal is hardest and potassium is softest among metals.

• Ductility :- It is the property of metals by which it can be stretched in length without breaking. For eg- Gold, Silver, Platinum can be easily drawn into wires.

• ~~Ma~~ Malleability :- It is the property by virtue of which a metal can be rolled into thin sheets without breaking. For eg- Gold, Silver, Aluminium, Copper etc.

• Toughness :- It is the property of a metal to resist repeated shocks or vibrations without breaking. For eg- Gold & Silver are tough metals.

- Tenacity :- Tenacity of a metal is measured by the weight which its wire support. For eg-

- Tensile Strength :- The tensile strength of a metal is the ability to carry a load without breaking. For eg-

- Weldability :- It is the process of ~~Using~~ Uniting two pieces of metal by means of heat by bringing their ends in the molten state. For eg-

- Soldering :- A method of joining the metal surfaces by introducing a molten — . ferrous alloy with melting point below ~~40~~ 400°C between them is known as Soldering.

- Machinability :- It is the property of metal which allows it to give the required shape with the help of machine.

- Forging :- Heating of a metal stock to a desired temparature enable it to acquire sufficient plasticity, followed by the operations like hammering, bending, pressing etc to give it the desired shape. This is known as forging.

- Extrusion :- When a piece of metal is forced to flow through a die orifice under pressure, it reduces its area of cross section. This process is known as extrusion.

- Castability :- It is the ability of a molten material to take exact dimensions of the mould.

- Brittleness :- Brittleness is the property of breaking on an impact. It is not the desirable property of an engineering material.

- Brazing :- A method of joining metal surfaces by introducing molten non ferrous alloy with melting point above 400°C between them is known as Brazing.

Q What do you mean by processing of Ore?

Ans - Ores are crushed in crushers and then converted into fine powder. It is the passed through sieves to get fine dust like powder. This is called as processing of Ore.

Q